mirafra
TECHNOLOGIES

N Satheesh Kumar
Experience: 1.5 years

Experience Summary

Now: Mirafra Software Technologies, Hyderabad, Software Engineer I , 17 th Mar 2026-Present.
Past: Hoags Technologies, Trainee Software Developer, May 2025 – Jul 2025
Past: Vector India, Embedded Trainee , Mar 2024 – Nov 2024

SKILLS

- **Programming Languages**: C, Data structures,Linux Device Driver Development, Embedded C, Data Structures using C, C++, Python.
- **Architectures:**good understanding on ARM, RISC-V,x86.
- **Operating Systems**: Linux, RTOS
- **Microcontrollers/SOCs :** LPC2129(ARM7 TDMI-S core), Raspberry Pi 4(BCM2711)
- **Bus Protocols:** UART, I2C, SPI,USB,PCIe CAN.
- **OS Concepts:**Cache Hierarchy, Scheduling, Process management, Virtual Memory.
- **Tools** : Keil µVision, Proteus.
- **Debugging Tools:**JTAG, SWD Linux Kernel Debugging
- **AI Tools**: Claud and Cursor.
- **Core Concepts:** Firmware Development,Linux Bootup Process, U-Boot
- **Version Control tools:**Git/GitHub, VS code

Linux Kernel and Driver Development:

- Linux Device Drivers: Strong knowledge of character device drivers (file operations, IOCTL handling, data transfer).
- Platform Drivers(GPIO, UART, I2C) and Device Tree configuration/integration.
- Interrupt Management: Skilled in implementing interrupt handlers with bottom-half mechanisms like tasklets, workqueues, and kernel threads.
- System Reliability: Exposure to kernel crash dump analysis (Oops, panics) and identifying memory leaks.
- Memory Handling: Practical understanding of kernel memory allocation techniques including kmalloc, vmalloc.

PROJECTS

Secure Real-Time Environment Monitoring System with Kernel-Level Access Control and Remote Alerts
Role: Embedded Linux Kernel Developer.
Responsibilities & Contributions:

- Designed and developed a Linux kernel-space environment monitoring system on Raspberry Pi and x86 platforms using custom character device drivers.
- Interfaced MCP3208, LM35 and LDR sensors using SPI and GPIO subsystems.
- Used workqueues and wait queues to defer processing from interrupt context and synchronize user-space data access.
- Developed buffered sensor data handling and asynchronous event processing for efficient kernel-space communication.
- Performed low level programming using ioremap(), readl() and writel().
- Enabled data logging and monitoring on Ubuntu-based systems through UART communication and kernel logs.
- Built and tested custom Linux kernel modules using cross-compilation and embedded Linux toolchains.
- Developed real-time threshold detection and alert generation for abnormal temperature and light conditions using kernel-space event handling.

Tools & Technologies Used: C Programming, Linux Device Drivers, UART, SPI, PWM, Kernel Threads Raspberry Pi 4 Model B, MCP3208, LM35, x86 Systems.

Linux Shell Implementation

- Implemented a custom shell in such a way that it supports basic Linux commands to handle file operations Pipelining, Signal Handling, etc.
- Built using Linux system calls, Signal handling, File management and maintained source code using git.
- Programmed in Embedded C using Keil µVision with interrupts, timers, and modular design for scalability and fault tolerance.
- Gained hands on experience with Linux internals and Process management.

Tools & Technologies: C programming, git, Makefile .,Linux System Programming.

Vehicle Dashboard Implementation using CAN

- Designed and developed a CAN-based automotive dashboard system using LPC2129 for real-time vehicle parameter monitoring.
- Developed low-level CAN driver initialization and message handling using register-level programming.
- Integrated real-time display of vehicle parameters such as speed, temperature, fuel level and warning indicators.
- Gained hands-on experience on Low level firmware development and CAN protocol and Error checking mechanisms.

Tools & Technologies: C programming, Keil, Embedded-C, LED, 16x2 Display, Flash Magic.

Student Data Base Management System

- Implemented a Dynamic Student data base management system
- Implemented CRUD operations (Add, Update, Delete, Search, Display) using dynamic memory allocation and pointer manipulation.
- Applied concepts of Data Structures, linked lists and file handling to manage student records through a menu-driven command-line interface.

Tools & Technologies: C programming, Singly Linked List, Dynamic Memory Allocation,GCC, Makefile, Bash.

ACADEMIC CREDENTIALS

Bachelor's in electrical and Electronics Engineering CGPA:8.43 — **Aug 2024**
Sri Venkateswara College of Engineering and Technology, Chittoor

Intermediate(MPC) | CGPA: 8.29 — **Jun 2020**
Mother Theresa Junior College, Palamaner

Secondary Education CGPA: 9.2 — **Mar 2018**
Zillaparishad High School, Bandlapalli